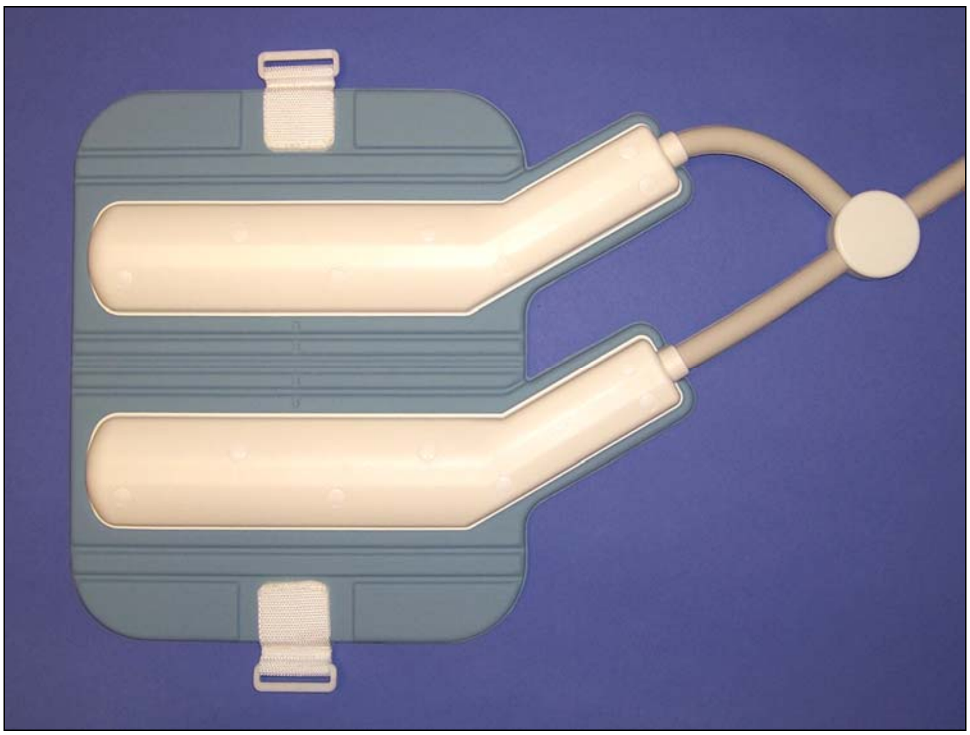
Operator Manual

1.5T & 3T GEM Small Anterior Array



Manufacturer

Invivo

Invivo Corporation
3545 SW 47th Avenue
Gainesville, FL 32608-7691 U.S.A.
(352) 336-0010

4535-303-05141 Rev 01

CE 0123

Distributed by:

GE Healthcare

First Edition
November 2010

Licenses and Trademarks
The Invivo Logo is a registered trademark of Invivo Corporation.

Proper performance of this coil is guaranteed only while the coil is being used on the MR system (hardware/software level) specified at the time of purchase. Upgrades or other modifications to the system software and/or hardware may affect compatibility. Prior to upgrading your MR system, please contact your GE Healthcare representative to discuss coil compatibility issues. Failure to do so may void your warranty.

Medical Device Directive
These products conform with the requirements of council directive 93/42/EEC concerning medical devices, when they bear the following CE Mark of Conformity:

CE 0123

EC REP
GBM Authorised Representative Ltd.
The White House, 2 Meadrow
Godalming, Surrey GU7 3HN
United Kingdom

Attention, Consult Accompanying Documents

Type BF Equipment

Class II Equipment

NOTICE:
This equipment shall be transported and stored under the following conditions:

Shipping
1. Ambient temperature range of -40°C to +65°C.
2. Relative humidity range of 10% to 100%, non-condensing.

Storage
1. Ambient temperature range of +10°C to +40°C.
2. Relative humidity range of 10% to 90%, non-condensing.

Atmospheric pressure range of 500 hPa to 1060 hPa.

WARNING: This product contains chemicals, including lead, known to the state of California to cause birth defects or other reproductive harm. **Wash hands after handling.**

Caution:
Federal law restricts this device to sale, distribution and use by or on the order of a physician.

INTRODUCTION

This manual describes the safety precautions, features, use and care of the Invivo 1.5T & 3T GEM Small Anterior Array, for use with the GE 1.5T MR450W GEM System or the GE 3.0T MR750W GEM System. The Invivo GEM Small Anterior Array is a Receive only coil. Please review this manual thoroughly before using the device.

The previous name for this coil was 32 Channel Cardiac Coil Anterior Array.

If you have any questions or comments regarding this manual, or need any assistance with the use of the product, please contact your GE Healthcare representative.

COMPATIBILITY

The Invivo 1.5T GEM Small Anterior Array, Invivo model number 9896-031-12151, is compatible with the GE 1.5T MR450W GEM System with P connector.

The Invivo 3.0T GEM Small Anterior Array, Invivo model number 9896-031-12161, is compatible with the GE 3.0T MR750W GEM System with P connector.

EXPLANATION OF SYMBOLS

Medical Device Directive – These products conform with the requirements of council directive 93/42/EEC concerning medical devices when they bear the CE Mark of Conformity.

Potential Hazard – To avoid harm, all safety messages that follow this symbol must be obeyed.

Type BF Equipment

Class II Equipment

Do not cross or loop cables. Arcing and patient burns could result.

Waste in Electrical and Electronic Equipment

A hazard exists which could cause serious injury.

MR Safe

Attention – Consult Accompanying Documents

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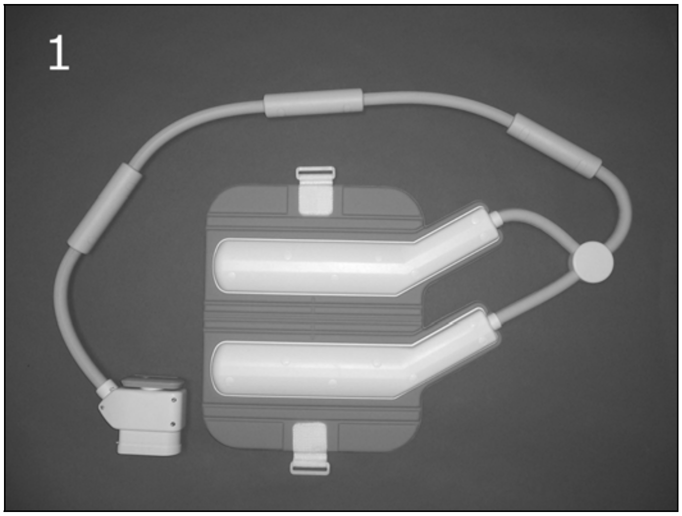
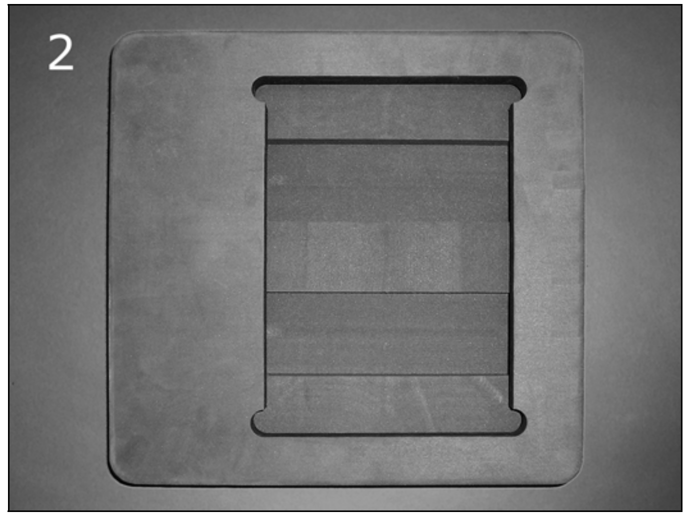
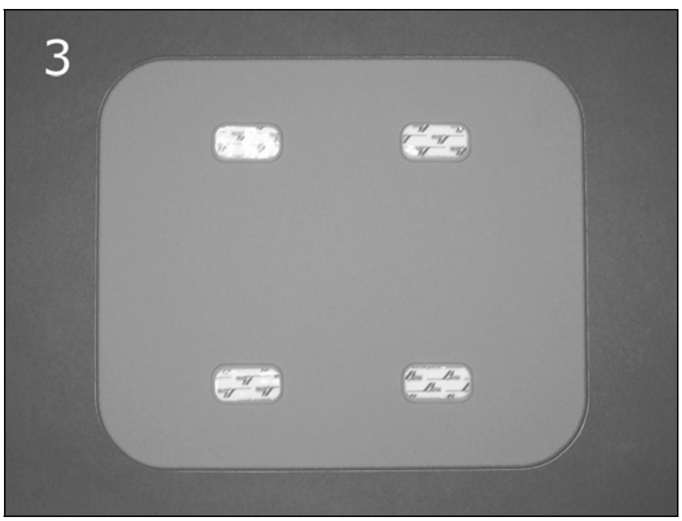
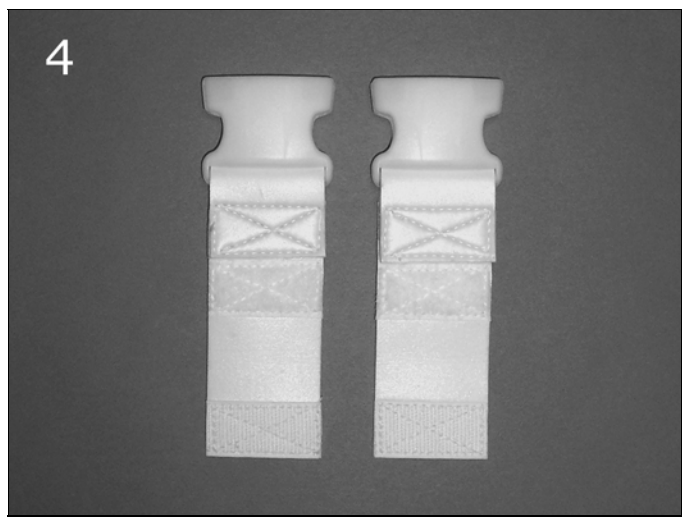
CHAPTER 1 – 1.5T & 3T GEM SMALL ANTERIOR ARRAY CONTENTS

The 1.5T & 3T GEM Small Anterior Array consists of the following parts. Please inspect upon receipt to make sure all parts have arrived and are in good order. Use this guide to refer to part names throughout this manual.









	Description	Invivo Part #	GE Part #	Qty
1	1.5T GEM Small Anterior Array	4535-302-48071	5366664-2	1
	3.0T GEM Small Anterior Array	4535-302-48111	5366627-2	1
2	Phantom Positioner (2 pieces)	4535-303-03571	5406404-11	1
3	Anterior Pad	4535-302-60711	5406404-10	1
4	Strap Set	4535-303-03591	5406404-12	1
5	Operator Manual	4535-303-05151	N/A	1

CHAPTER 2 – SAFETY

2-1 Training

This manual contains detailed information regarding the set-up, positioning and use of the Invivo Corporation coil. The instructions must be read carefully and thoroughly before attempting to scan patients with this coil.

2-2 Quality Assurance

The procedure described in the Quality Assurance chapter of this manual should be performed upon receipt of the coil to establish a baseline of coil performance. The procedure should be repeated at regular intervals.

2-3 Indications

The coil is indicated for use, on the order of a physician, in conjunction with an MR scanner as an accessory to produce images of the cardiovascular, pulmonary, renal, and abdominal systems, as an aid to diagnosis.

2-4 Contraindications

The operator should be aware of the following contraindications for use related to the strong magnetic field of the MR system:

⚠ Scanning is contraindicated for patients who have electrically, magnetically or mechanically activated implants (cardiac pacemakers, for example). The magnetic and electromagnetic fields produced by the MR system and coil may interfere with the operations of these devices.

⚠ Scanning patients with intracranial aneurysm clips is contraindicated.

2-5 Precautions

Precautions should be taken when scanning patients with the following conditions:

⚠ Greater than normal potential for cardiac arrest

⚠ An increased likelihood for developing seizures or claustrophobia

⚠ Unconscious, heavily sedated, or confused patients

⚠ Inability to maintain reliable communications

2-6 Cautions

The following general warning statements apply to scanning with a magnetic resonance system. For further details, review the warnings in your MR system Operator's Manual.

Do not cross or loop cables. Arcing and patient burns could result. Route cables away from patient so that they do not touch the patient.

Assure that the patient is not touching the bore. If necessary, place pads between the patient and the surface of the bore.

If the patient complains of warming, tingling, stinging, or similar sensations, promptly stop the scan procedure, examine the patient, and contact the responsible physician before continuing the procedure. Pay special attention to very young, sedated, or other compromised patients who may not be able to communicate effectively.

Patients with implanted medical devices/components should not be scanned because the magnetic field may interact with the medical device/component. Consult patient's physician

Persons with cardiac pacemakers or other implanted electronic devices should not enter the magnetic field zone delineated by the MR system manufacturer.

There is a risk to scanning feverish or decompensated cardiac patients.

Do not allow the patient to bring magnetic objects or non-magnetic metallic objects – such as jewelry, hairpins, prosthetics or clothing containing metal components – into the magnet. Such objects will interfere with the imaging capabilities of the system and coil.

Certain transdermal patches may cause burns to the underlying skin due to absorption of RF energy. The supplier of the patches should be consulted or the patch should be removed to avoid burns. A new patch should be applied after the examination.

Facial makeup should be removed before scanning because it may contain metal flakes which can cause skin and eye irritation. Permanent eyeliner tattoos may cause eye irritation due to ferromagnetic particles.

Patients who work in environments in which there is a risk of having embedded metallic fragments in or near the eye should be carefully screened before undergoing a MR exam.

⚠ Visually inspect the cable insulator jackets, strain reliefs, and connector boxes before each use. If the insulation is broken, or if the cable is frayed, immediately discontinue use of the device.

⚠ Ensure that the patient does not form a loop with any body parts. For breast coils in particular, patients are imaged lying prone with both arms extended above the head (superman). **Do not allow patients to touch any part of the right hand or arm to any part of the left hand or arm. The loop formed by doing so could cause an RF burn at the point of contact. Use pads to ensure the patient's hands do not touch any metal features of the patient table or cradle**. Be sure to educate the patient accordingly and check the patient's position immediately before the scan.

2-7 Emergency Procedures

In the unlikely event that a coil creates smoke, sparks or makes an unusually loud noise, or if the patient requires emergency assistance:

- Stop the scan if one is in progress.
- Remove the patient from the scan room if medical treatment is needed.

2-8 Technical Considerations

⚠ The coil and accessories require special conditions regarding electromagnetic compatibility. The coil must be installed and used in a shielded scan room provided with the MR magnet and system. The user must ensure that the scan room door is closed during system use. Failure to do so may cause reciprocal interference with portable and mobile RF communications equipment, affecting the performance of the MR coil and/or such equipment.

⚠ The coil should only be used with the cable and accessories specified in the system operator manual.

⚠ The use of accessories other than those specified in the system operator manual may result in decreased ESD immunity of the coil or MR system, causing damage to the coil and/or system.

⚠ The equipment should not be used with other coils or equipment present in the MR scanner except as specified in the system operator manual.

⚠ Tampering with the cable pins and connector may damage the connector and affect coil or system performance. Please verify that connector and pins are not damaged before use.

⚠ The coil should not be left unplugged in the system during body coil scanning.

CHAPTER 3 – QUALITY ASSURANCE

To perform a Quality Assurance test, please call your GE Service Representative. Or, refer to the appropriate System Service Manual for the most up to date procedure, and follow the instructions for Coil Imaging Performance Verification. System Service manuals can be found on the class A Service Methods CD shipped with the system, under this navigation path: Installations -> Coils -> 1.5T (or 3T) Coil Vendor Manuals.

CHAPTER 4 – USING THE GEM SMALL ANTERIOR ARRAY

4-1 Connecting the Cable

The GEM Small Anterior Array must be connected to the P Port connector.

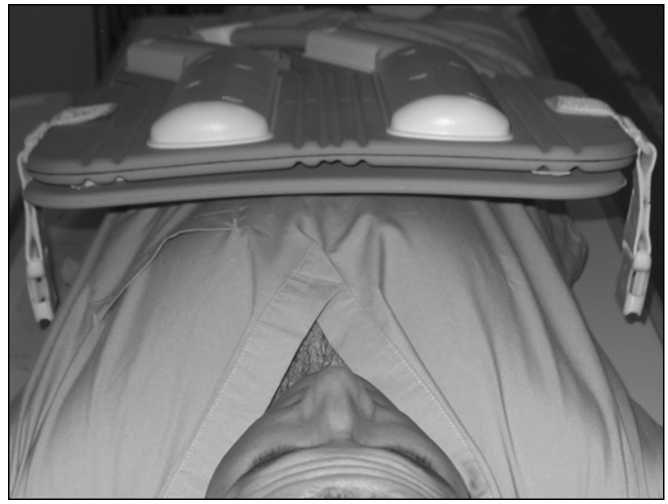
4-2 Positioning the Patient

Position the patient on the patient cradle.

4-3 Positioning The Coil

Position the GEM Small Anterior Array, along with the anterior pad, on the patient. The anterior pad is supplied with the coil, and goes between the patient and the coil.

Ensure the GEM Small Anterior Array is centered right to left on the patient.



Use the Velcro straps to secure the GEM Small Anterior Array on the patient. The Velcro straps are supplied with the coil.

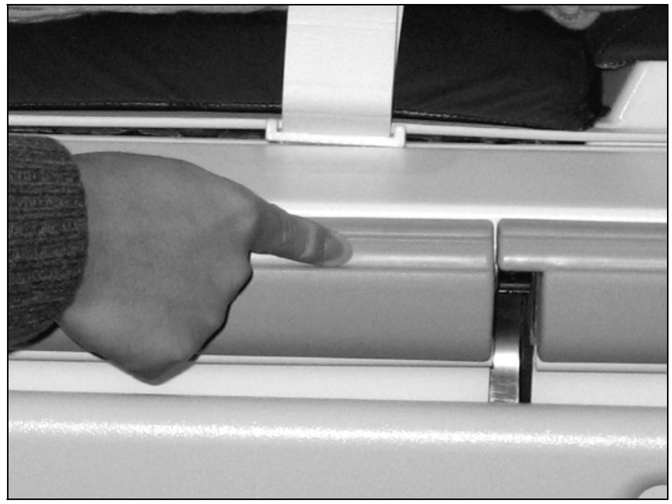
⚠ Ensure that the patient does not form a loop with any body parts. For breast coils in particular, patients are imaged lying prone with both arms extended above the head (superman). **Do not allow patients to touch any part of the right hand or arm to any part of the left hand or arm. The loop formed by doing so could cause an RF burn at the point of contact. Use pads to ensure the patient's hands do not touch any metal features of the patient table or cradle**. Be sure to educate the patient accordingly and check the patient's position immediately before the scan.

4-4 Cardiac AA Indexing & Patient Landmark

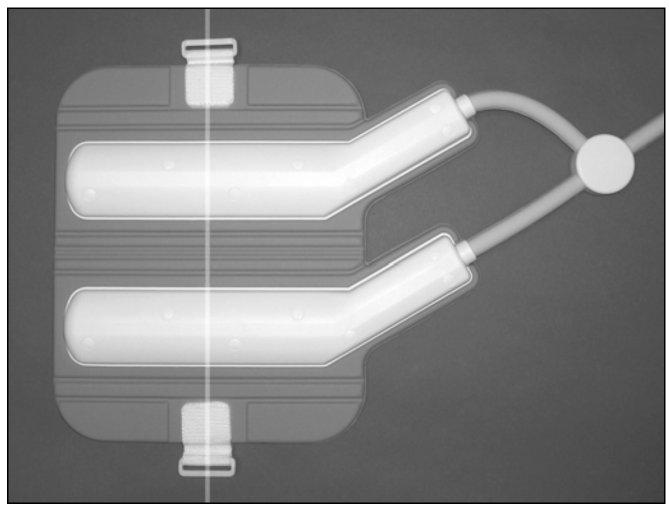
1. Move the table to the home position.

2. Press the IntelliTouch strip at the location aligned with the Cardiac Anterior Landmark line.

   The IntelliTouch strip is located at the table edge under the rubber cover. This step tells the system the location of the GEM Small Anterior Array.



3. Landmark the coil on the center Landmark, and advance the patient into the magnet for scanning.



4-5 Field of View and Coverage

For heart studies an FOV of 35 cm L/R is suggested, depending upon patient size. A smaller FOV of 27 cm S/I may be used if desired.

4-6 Patient Hearing Protection

Provide ear plugs for the patient after all instructions have been given.

WARNING!

Hearing protection is required for all people in the scan room during a scan to prevent hearing impairment. Acoustic levels may exceed 99 dB(A).

Hearing protection must have a Noise Reduction Rating (NRR) of 28 dB or better (e.g., 30 dB, 32 dB, etc.).

CHAPTER 5 – COIL MODE CONFIGURATION

The GEM Small Anterior Array has two modes of operation:

<table>
<tr><th>Coil Name</th><th>Mode Name</th></tr>
<tr><td>GEM AA only</td><td>Small AA only 30</td></tr>
<tr><td>GEM Whole Body</td><td rowspan="2">Body 30 Small</td></tr>
<tr><td>GEM Chest/Body/Pelvis</td></tr>
</table>

CHAPTER 6 – SCANNING

6-1 Auto Pre-scan

Generally, the system auto pre-scan will function normally when used with the GEM Small Anterior Array. If auto pre-scan returns the message "auto pre-scan failed to complete normally," then manual pre-scan may be required to acquire the image. Consult the system operator's manual for manual pre-scan procedures.

6-2 Localizing with the Body Coil

The system body coil may be used at any time while the GEM Small Anterior Array is in the scanner and connected to the system port. This allows a large FOV (45 cm.) body coil localizer to be performed, if desired.

6-3 Using Autoshim

AUTOSHIM is a feature of the AUTOPRESCAN software that improves image quality. It does this by improving the magnetic field homogeneity within the FOV selected. The improvement in image quality is often dramatic when the selected FOV is far off center, and when acquiring FSE or T2 weighted scans, or FAT SAT images or when acquiring cardiac images on 3T systems. 10-15cm shimming volume should be placed at the region of interest for optimal image quality.

6-4 Protocols

Use the system recommended protocols for scanning with this coil for optimum results and best performance.

Specific choice of protocols is entirely the responsibility of the radiologist and/or technologist.

CHAPTER 7 – INSTALLATION AND MAINTENANCE

7-1 Installation and Configuration

Coil installation and configuration requires the services of your GE Service Representative. Please refer to the Service Manual for coil installation and configuration information.

7-2 Cleaning

The GEM Small Anterior Array and pads must be cleaned after each use and stored using the following procedure:

- Wipe with a cloth that has been dampened in a solution of 10% bleach and 90% tap water, or 30% isopropyl alcohol and 70% tap water.
- **Do not pour any cleaning solution directly on the coil!**
- Let the coil housing and pads dry before use.
- **Under no circumstances should the coil be placed into any type of sterilizer.**

7-3 Storage

This equipment must be stored under the following conditions:

- Ambient temperature range of +10°C to +40°C
- Relative humidity range of 10% To 90%, non-condensing
- Atmospheric pressure range of 500 hPa TO 1060 hPa

To store the coil, a storage space of 48.3 cm/19.0 in (width) x 16.5 cm/6.5 in (depth) x 101.6 cm/40.0 in (length) is required.

7-4 Replaceable Accessories

The following table lists replaceable accessories that may be purchased from Invivo Corporation:

1.5T GEM Small Anterior Array

Description	Invivo #	GE #
Anterior Pad	4535-303-03611	5406404-10
Velcro Strap kit	4535-303-03621	5406404-12
Phantom Positioner	4535-303-03601	5406404-11
Anterior Coil	4535-302-86221	5366664-2

3.0T GEM Small Anterior Array

Description	Invivo #	GE #
Anterior Pad	4535-303-03611	5406404-10
Velcro Strap kit	4535-303-03621	5406404-12
Phantom Positioner	4535-303-03601	5406404-11
Anterior Coil	4535-302-86241	5366627-2

7-5 Contact Information

For additional information, contact:

GE Healthcare Americas
USA ....…......... 800-558-5102
Canada ......….. 800-668-0732

GE Healthcare Asia
China ....…... 86-21-62192228
Taiwan ….... 886-2-2505-7900
Singapore …...... 65-291-8528
Australia ….... 61-2-9975-5501
Japan ...…..... 81-120-48-2630
Korea ..…...... 82-31-740-6119
India ..…........ 91-80-845-2923

GE Healthcare Europe
(33) 1-41-19-76-76

GE Healthcare Web Site
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